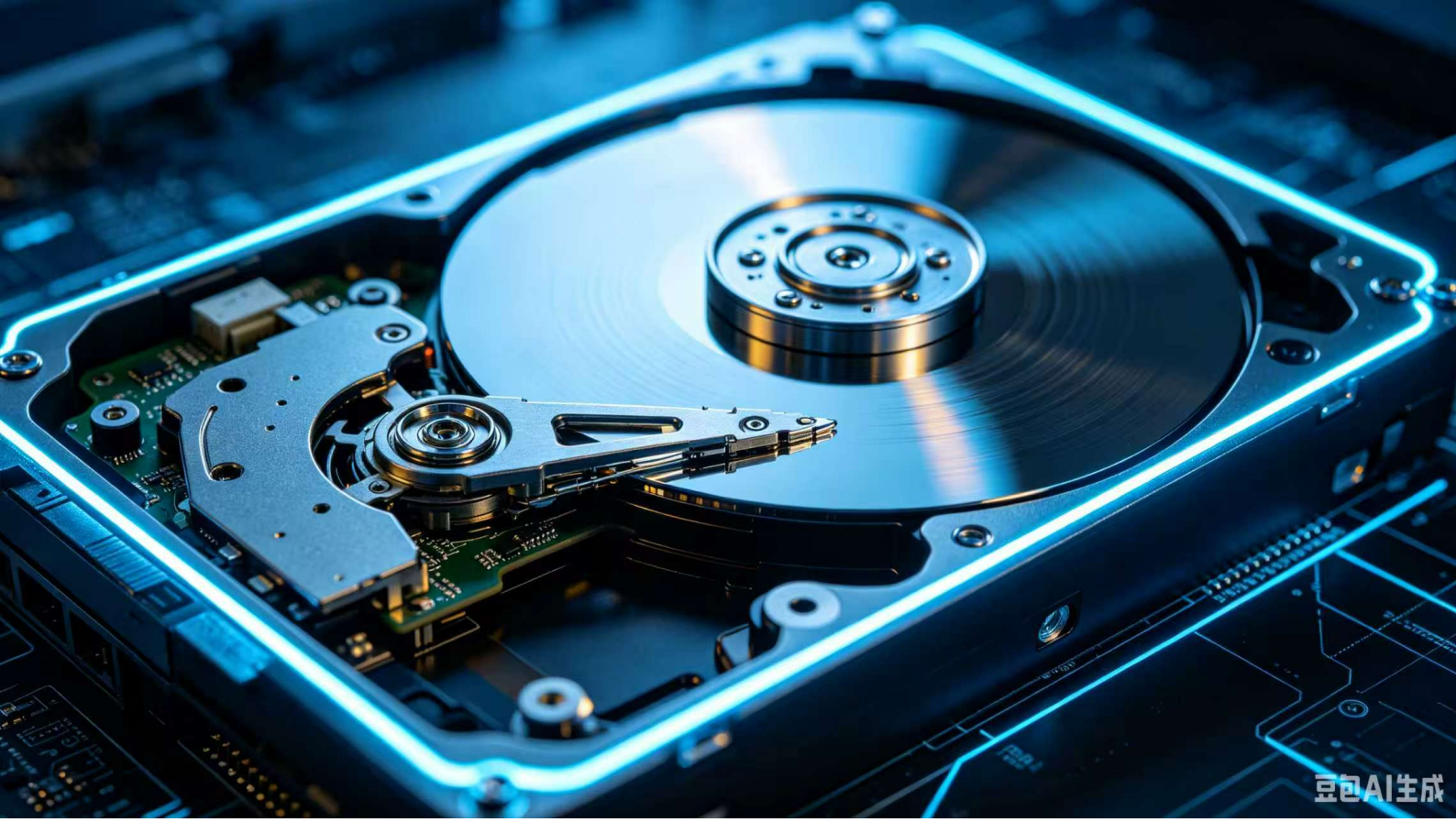


8. The Magnetic Field

> In order to obtain physical ideas without adopting a physical theory we must make ourselves familiar with the existence of physical analogies.
>
> – *J. C. Maxwell*

8.1 Charged Particles in a Magnetic Field

In addition to the electric field, there is also a **magnetic field**. Magnetic fields are produced by magnets and by electric currents. Similar to the electric field, we can also represent magnetic fields with field lines. Thus, the tangential direction of the field line at a specific point gives the direction of the magnetic field **B**; and the density of the field lines at a specific point is proportional to the magnitude of **B**. However, the magnetic field lines have no beginning and no end. This is drastically different from static electric field lines, which always start or end at a charge, or run to infinity. This property of the magnetic field lines reflects the fact that no *magnetic monopole*, or magnetic charge, has ever been observed. All magnets have two poles, i.e. a north pole and a south pole. Hence they are also called **magnetic dipoles**. Outside the magnet, the magnetic field lines emerge from the north pole and enter at the south pole; while inside the magnet, the field lines point from the south pole towards the north pole.

It is an experimental observation that charged particles can deviate from their straight paths in a magnetic field. The force exerted by the magnetic field on a charged particle is called the **Lorentz force**, given by:

$$\mathbf{F}_B = q\mathbf{v} \times \mathbf{B}$$

where $\mathbf{v}$ is the velocity of the charged particle, and $\mathbf{B}$ is the magnetic field. The Lorentz force is always perpendicular to both $\mathbf{v}$ and $\mathbf{B}$. Since $\mathbf{F}_B \perp \mathbf{v}$, the speed is unchanged, and the work done by the magnetic force on a single charged particle is zero.

The SI unit for $\mathbf{B}$ is the tesla (T):

$$1\ \mathrm{T} = 1\frac{\mathrm{N}}{\mathrm{C} \cdot (\mathrm{m/s})} = 1\frac{\mathrm{N}}{\mathrm{A} \cdot \mathrm{m}}.$$

An earlier (non-SI) unit still in common use is the gauss (G), with $1\ \mathrm{T} = 10^4$ G. Earth's magnetic field near the planet's surface is about 0.5 G ($\approx 50\ \mu$T).

Suppose now that a particle with charge q and mass m passes through a uniform magnetic field $\mathbf{B}$. In general the velocity of the particle $\mathbf{v}$ can be decomposed into $\mathbf{v}_\parallel$ that is parallel to $\mathbf{B}$ and $\mathbf{v}_\perp$. If there is no other field in the space, the Lorentz force $\mathbf{F}$ is the only force acting on the particle. Therefore, $\mathbf{v}_\parallel$ will remain constant, and $\mathbf{v}_\perp$ will change its direction but its magnitude will also remain constant. Thus even without explicit calculations, we conclude that the motion of the particle must be circular with constant angular speed ω in the plane that is perpendicular to $\mathbf{B}$, while at constant speed $v_\parallel$ in the direction of $\mathbf{B}$. This is a *helical* motion, of which a circle ($v_\parallel = 0$) and a straight line ($v_\perp = 0$) are the limiting cases.

Mathematically, $F = |q|v_\perp B$ and its direction is perpendicular to both $\mathbf{B}$ and $\mathbf{v}_\perp$. Such a force provides the centripetal force of the circular motion, therefore:

$$|q|v_\perp B = \frac{mv_\perp^2}{R}$$

where R is the radius of the circular motion. Therefore, noticing that $v_\perp = \omega R$, we have:

$$R = \frac{mv_\perp}{|q|B}, \qquad \omega = \frac{|q|}{m}B$$

The quantity $\frac{|q|}{m}$ is an intrinsic property of the particle, called the *charge-mass ratio*. The sense of the circular motion is opposite for the two signs of charge: as a vector, $\boldsymbol{\omega} = -q\mathbf{B}/m$.

8.1.1 Cyclotron

A cyclotron uses a magnetic field to repeatedly bring charged particles back to an accelerating region. The device consists of two hollow D-shaped electrodes ("dees") with a gap between them. An alternating electric field in the gap accelerates particles each time they cross it, while the perpendicular magnetic field bends them into semicircular paths. The cyclotron angular frequency is

$$\omega = \frac{|q|B}{m},$$

which is independent of the particle's speed (for non-relativistic velocities). The electric field in the gap must alternate at the cyclotron frequency $f = |q|B/(2\pi m)$, so that particles always experience an accelerating field when crossing the gap.

Example 8.1: Motion of a charged particle in space with both an electric and a magnetic field.

In the general case, when there is both a magnetic and an electric field in space, the motion of the charged particle could be complicated. For simplicity, we shall assume that both fields are uniform, meaning $\mathbf{E}$ and $\mathbf{B}$ are constants. We consider a charged particle with charge q and mass m. Thus, the force experienced by this particle is:

$$\mathbf{F} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}$$

Let's consider two simple cases:
Case 1: $\mathbf{E}$ and $\mathbf{B}$ are in parallel (Figure 8.1, left panel).
Let's assume both of them are in the $\hat{z}$-direction. Thus the Lorentz force lies in the xy-plane, while the Coulomb force gives the particle a uniform acceleration along $\hat{z}$. The Lorentz force leads to its uniform circular motion in the xy-plane. Mathematically, we can write down:

$$F_z = qE = m\frac{d^2z}{dt^2}$$

$$F_x\hat{x} + F_y\hat{y} = q(\hat{x}v_yB - \hat{y}v_xB) = m(a_x\hat{x} + a_y\hat{y})$$

The acceleration along $\hat{z}$ is $a = qE/m$, and the angular speed in the xy-plane is $\omega = |q|B/m$.
Case 2: $\mathbf{E}$ perpendicular to $\mathbf{B}$. Initial velocity $\mathbf{v} = 0$ (Figure 8.1, right panel).
Let's assume that $\mathbf{E} = E\hat{x}$ and $\mathbf{B} = B\hat{z}$. Thus, the Lorentz force will be perpendicular to $\hat{z}$ and the Coulomb force will be in $\hat{x}$, therefore the total force is in xy-plane, and the motion is also within the xy-plane. Assuming that $\mathbf{v} = v_x\hat{x} + v_y\hat{y}$ at a specific time, the equation of motion will be:

$$qE + qv_yB = m\frac{dv_x}{dt}$$

$$-qv_xB = m\frac{dv_y}{dt}$$

Rewriting the above equations using $\omega = qB/m$ and $a = qE/m$, we have:

$$a + \omega v_y = \frac{dv_x}{dt}$$

$$-\omega v_x = \frac{dv_y}{dt}$$

Differentiating the first equation once more and substituting the second into the result, we have an equation for v_x: $\ddot{v}_x + \omega^2 v_x = 0$. The solution is $v_x = v_0\sin(\omega t + \phi)$ and $v_y = v_0\cos(\omega t + \phi) - a/\omega$. Noticing the initial condition yields $\phi = 0$ and $v_0 = a/\omega = E/B$. The drift speed E/B is independent of both q and m: the drift velocity is $\mathbf{v}_d = \mathbf{E} \times \mathbf{B}/B^2$, a vector fixed by the fields alone.

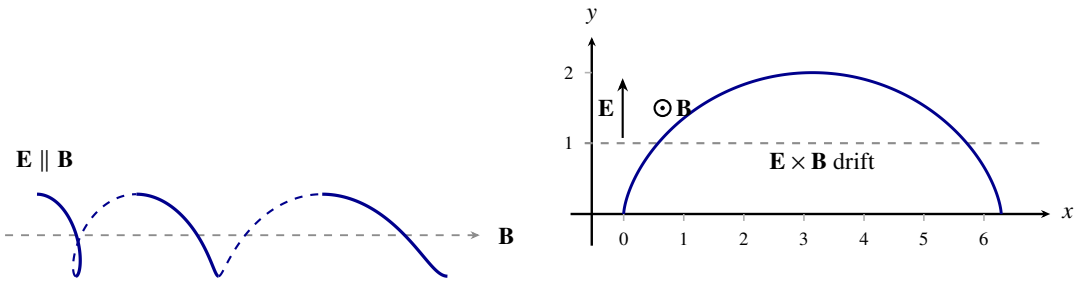


Figure 8.1: Motion of a charged particle in uniform electric and magnetic fields. Left: when $\mathbf{E} \parallel \mathbf{B}$, the particle follows a helical path with uniform acceleration along $\mathbf{B}$. Right: when $\mathbf{E} \perp \mathbf{B}$ with zero initial velocity, the trajectory is a cycloid; this panel takes $\mathbf{E} = E\hat{y}$ and $\mathbf{B} = B\hat{z}$, the transpose of the axes used in the text, so that here the bounded coordinate is y and the drift $\mathbf{v}_d = \mathbf{E} \times \mathbf{B}/B^2$ is along $+x$, the dashed line.

Of course, we can also apply another method to think about the last case. Imagine that we change our frame of reference from the original one to another one moving in the $-\hat{y}$ direction with speed $v_0 = E/B$. The reason for such a shift of frame is that, after it, a point charge at rest would remain at rest, meaning that the effect of the electric field can be "canceled" by such a choice of frame and the magnetic field. In other words, by replacing $\mathbf{v} = \mathbf{u} + \mathbf{v}_0$, the Lorentz force can be written as:

$$\mathbf{F}_B = q\mathbf{v} \times \mathbf{B} = q\mathbf{u} \times \mathbf{B} + q\mathbf{v}_0 \times \mathbf{B}$$

and the last term exactly cancels $\mathbf{F}_E = q\mathbf{E}$, thus the total force experienced by the point charge in the new reference frame is only $q\mathbf{u} \times \mathbf{B}$, or only the magnetic field exists in the new frame. This holds provided $E < cB$, which is satisfied in every case considered here: the substitution above is the non-relativistic version of the statement, and it uses the same B in both frames, which is accurate to order $(E/cB)^2$; the field in the new frame is in fact $B' = B\sqrt{1 - E^2/(c^2B^2)}$, smaller than B. Thus, the particle will be in circular motion with angular frequency $\omega = |q|B/m$ in the new frame. Thus, by shifting back to the original frame, we have exactly the previous solution.

8.1.2 Discovery of Electrons

In 1897, J. J. Thomson discovered an unknown charged particle by measuring its charge-mass ratio. He claimed this particle is lighter than the hydrogen atom by a factor of more than 1000, and that these particles can be found in all matter. These particles are now known as *electrons*. His experiment was conducted with a beam of cathode rays in a crossed electric and magnetic field, in which the two fields were adjusted until the beam was undeflected.

The electrons are emitted from a hot filament at the rear of the evacuated tube (Figure 8.2) and are accelerated by an applied potential difference V. After passing through a slit in the plate C, they form a narrow beam. Then, they pass through a region of crossed $\mathbf{E}$ and $\mathbf{B}$ fields and head toward a fluorescent screen S, where they produce a spot of light.

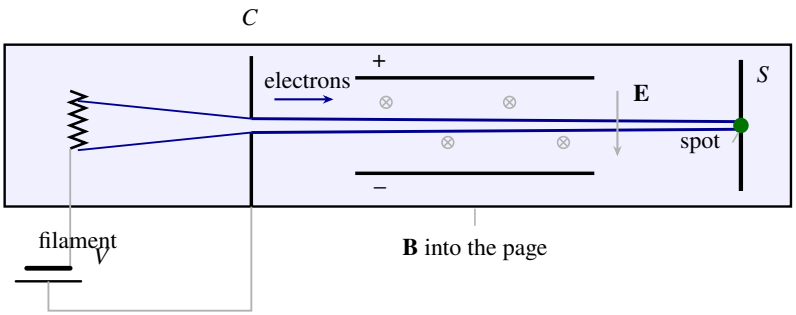


Figure 8.2: Thomson's apparatus for measuring the charge-mass ratio of the electron. Electrons from the hot filament are accelerated through a potential difference V and collimated by a slit in the plate C. In the region of crossed **E** and **B** fields the two forces balance when $E = vB$, so the beam is undeflected and strikes the center of the fluorescent screen S.

If the spot of light is located exactly at the center of the screen, then there must be constraints on V, B and E. The velocity of the electrons passing the slit can be evaluated by $v = \sqrt{\frac{2|q|V}{m}}$. Then, if the electrons are not deviated, the Coulomb force and the Lorentz force must cancel each other, giving $qE = qvB$, or $E = vB$. Combining with the expression for velocity, we have:

$$E = \sqrt{\frac{2|q|V}{m}}B$$

Thus, the charge-mass ratio:

$$\frac{|q|}{m} = \frac{1}{2V}\left(\frac{E}{B}\right)^2$$

8.1.3 The Hall Effect

Assume a strip of conductor of width d carrying a current i. Suppose there is a magnetic field perpendicular to the strip. Recall that the current density is $\mathbf{j} = nq\mathbf{v}_d$, with q the signed charge of a carrier, and the current is $i = |\mathbf{j}|A = n|q|v_d A$, where A is the cross-sectional area of this strip and n is the number density of charge carriers. The Lorentz force due to the magnetic field deflects the charge carriers to one side of the strip, and therefore a transverse voltage, the **Hall voltage** $V_H = Ed$, builds up across the width of the strip. Once the system reaches equilibrium, the electric and magnetic forces on a carrier are in balance, therefore:

$$\mathbf{E} + \mathbf{v}_d \times \mathbf{B} = \mathbf{0}$$

Substituting $\mathbf{v}_d = \mathbf{j}/(nq)$ gives

$$E = \frac{B}{nq}j$$

This looks similar to the definition of resistivity, except that now the current and the potential drop are not in the same direction, but perpendicular to each other. In addition, the expression has magnetic field B built-in. It is called **Hall resistivity**, defined as:

$$\rho_H = \frac{B}{nq}$$

and the Hall coefficient as:

$$R_H = \frac{1}{nq}$$

where q is signed. Thus the sign of R_H gives the sign of the charge carriers, and its magnitude gives their number density, provided that a single species of carriers conducts.

8.2 Magnetic Force on a Current-Carrying Wire

Suppose a straight, uniform wire of length L, carrying a current of $i = n|q|v_d A$, is put in a uniform magnetic field of $\mathbf{B}$. Each charge carrier of this wire experiences $\mathbf{F}_q = q\mathbf{v}_d \times \mathbf{B}$. There are nAL carriers in total, thus the total force acting on the wire is:

$$\mathbf{F} = q\mathbf{v}_d \times \mathbf{B}\, nAL = \mathbf{j} \times \mathbf{B}\, AL$$

The magnetic force above acts on the mobile carriers; collisions with the lattice, and the internal electric field that builds up when the carriers are pushed against the side of the wire, transmit that momentum to the wire as a whole, which is why the result is written as the force on the wire. Notice that the direction of $\mathbf{j}$ is indeed parallel to $\mathbf{L}$, and $i = |\mathbf{j}|A$, we can rewrite the above expression into:

$$\mathbf{F} = i\mathbf{L} \times \mathbf{B}$$

If the wire is not straight, or the field is not uniform, we can always break the wire into small pieces and write:

$$d\mathbf{F} = id\mathbf{l} \times \mathbf{B}$$

This force is called the **Ampère force**.

Question: Can Ampère force do work? Why? (See Exercise 9.)

8.3 Torque on a Current Loop

We now apply the previous results to a current loop. For simplicity, we consider a rectangular current loop of sides a and b in a uniform magnetic field $\mathbf{B}$, mounted so that it can turn about the axis through the midpoints of the sides of length b, an axis perpendicular to $\mathbf{B}$. Assuming that the current in the loop is i, the total force on the loop in this case is zero: the forces on the two sides of length a are equal and opposite, and so are those on the two sides of length b. The total torque, however, is not zero. Let θ be the angle between $\mathbf{B}$ and the normal to the loop. Each side of length a is perpendicular to $\mathbf{B}$, and so feels a force of magnitude iaB; the two are oppositely directed but act along different lines, and therefore form a couple. The forces on the two sides of length b are equal and opposite but directed along the axis itself, so they neither turn the loop nor move it. Their lever arm is $b \sin\theta$, so:

$$\tau = iabB \sin\theta = iAB \sin\theta, \qquad \boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$$

The scalar result above is for a single turn; a coil of N turns, each carrying the same current i, multiplies the torque by N. Here $\boldsymbol{\mu} = Ni\mathbf{A}$ is defined as the **magnetic dipole moment** of a coil with N turns and area $\mathbf{A}$ carrying a current i. The direction of $\mathbf{A}$ and the current direction follow the right-hand rule: if your four fingers follow the direction of i, the thumb points in the direction of $\mathbf{A}$. Although the calculation above is for a rectangle, both results, $\tau = iAB \sin\theta$ and $\boldsymbol{\tau} = \boldsymbol{\mu} \times \mathbf{B}$, hold for a plane loop of any shape.

8.3.1 Electric Motor

In a practical electric motor, the current in the rotating coil must be reversed each time the magnetic dipole moment $\boldsymbol{\mu}$ aligns with $\mathbf{B}$, so that the torque continues to rotate the coil rather than settling into equilibrium. This automatic reversal is achieved via a **split-ring commutator** that electrically connects the rotating coil with stationary contacts (brushes) supplying current from an external source (Figure 8.3).

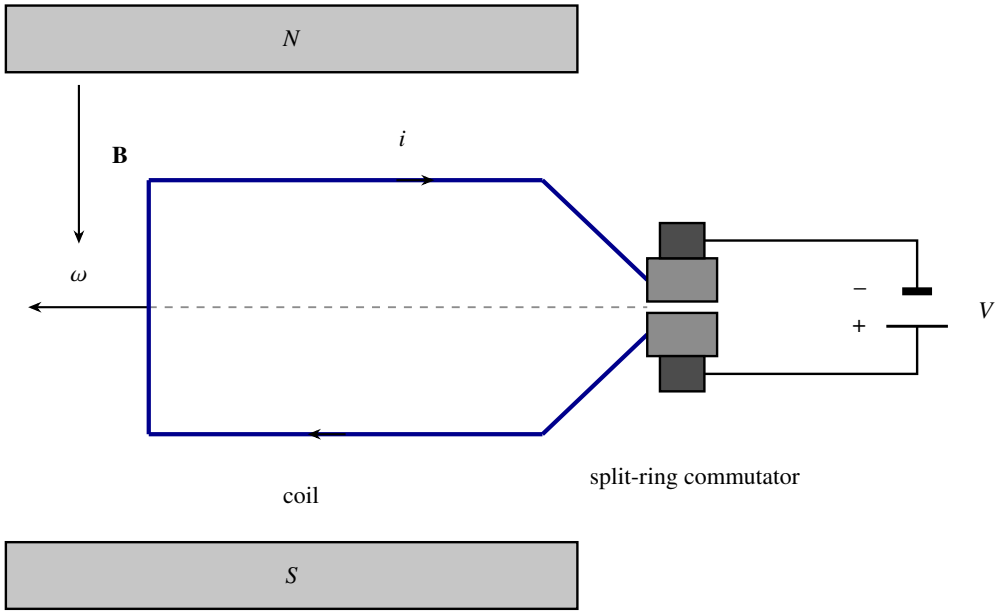


Figure 8.3: A direct-current electric motor, drawn at the instant the plane of the coil is the plane of the figure; the dashed line is the axis about which the coil turns. The coil turns in the field between the N and S pole faces; the split-ring commutator reverses the current in the coil every half turn, so the torque continues to act in the same sense.

8.4 Exercises

1. A particle with mass m and charge q enters a region of uniform magnetic field B large enough to contain at least one full turn of the helix. The initial velocity has magnitude v and makes an angle of 30° with the magnetic field direction; ignore gravity.

   (a) Describe the resulting motion (discuss separately the component of velocity parallel/perpendicular to the field).

   (b) Calculate the pitch (distance traveled along the field direction per revolution) of the helical path.

2. A thin copper strip with thickness $t = 0.50$ mm and width $w = 2.0$ cm carries a current of $I = 15$ A in the $+x$ direction. The strip is placed in a uniform magnetic field of magnitude $B = 1.5$ T pointing in the $+z$ direction (perpendicular to the flat face of the strip). A Hall voltage of magnitude $V_H = 3.3\,\mu$V is measured across the width of the strip. Assume that each carrier has a charge of magnitude e.

   (a) Calculate the drift velocity v_d of the charge carriers.

   (b) Determine the number density n (charge carriers per unit volume) in the copper.

3. A magnetic field B along the $+z$ axis is present in the $y \geq a$ region of the xy-plane. If we put a circular loop of radius R ($0 \leq a < R$) and current I (counterclockwise seen from $+z$) in the xy plane with its center located at the origin, what would the total force be on the loop?

4. A plastic disk is rotating with angular velocity ω in a uniform magnetic field B that is perpendicular to the disk's rotating axis (Figure 8.4). The disk has radius R, and it has a uniform surface charge density σ. Compute the magnitude of the total torque exerted by the magnetic field on the disk.

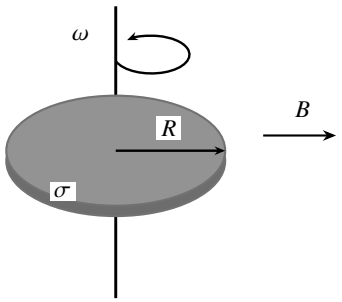


Figure 8.4: A rotating plastic disk in a uniform magnetic field. The disk turns about the axis shown, with angular velocity ω; its surface charge density is σ and its radius R.

5. In a cyclotron the two dees have radius R and lie in a uniform magnetic field of magnitude B perpendicular to the plane of the orbits. A particle of mass m and charge $q > 0$ starts from rest at the center and is accelerated across the gap between the dees every time it crosses, by a potential difference of magnitude V. Neglect the time spent in the gap and any relativistic correction.

   (a) Find the kinetic energy of the particle when its orbit has grown to radius R, and show that it does not depend on V.

(b) Find the number of gap crossings needed to reach that energy, assuming V is such that this number comes out whole. Say in one line what limits the energy the machine delivers when it does not.

6. Uniform, time-independent fields $\mathbf{E} = E\hat{z}$ and $\mathbf{B} = B\hat{z}$, with $E > 0$ and $B > 0$, fill all of space. At $t = 0$ a particle of mass m and charge $q > 0$ is at the origin with velocity $\mathbf{v}(0) = v_\perp \hat{x} - u\hat{z}$, where $v_\perp > 0$ and $u > 0$. Ignore gravity.

   (a) Find the position and the velocity of the particle at the first time $t_* > 0$ at which it returns to the plane $z = 0$.

   (b) Find the values of E for which it returns to the origin itself at that moment.

7. Uniform fields $\mathbf{E} = E\hat{x}$ and $\mathbf{B} = B\hat{z}$, with $E > 0$ and $B > 0$, fill all of space. At $t = 0$ a particle of mass m and charge $q \neq 0$ is released from rest at the origin; q may be of either sign. Ignore gravity.

   (a) Find the average velocity of the particle over one cyclotron period $T = 2\pi m/(|q|B)$, and show that it is the same for every q and every m.

   (b) Find the largest value of $|x|$ that the particle reaches, and state for which sign of q it moves toward $+\hat{x}$.

8. A beam of positive ions of charge $q > 0$, with a spread of speeds, travels along $-\hat{y}$ into a velocity selector, in which a uniform electric field $\mathbf{E}_1 = E_1\hat{x}$ and a uniform magnetic field $\mathbf{B}_1 = B_1\hat{z}$ act together, with $E_1 > 0$ and $B_1 > 0$. The ions that cross it undeflected pass through a slit at the origin into a second region, where the only field is a uniform $\mathbf{B}_2 = B_2\hat{z}$ with $B_2 > 0$, filling a region large enough to hold the whole half circle. There each ion turns through a half circle and strikes a detector occupying the part $x < 0$ of the plane $y = 0$.

   (a) Find the speed the selector transmits, and show that it depends on neither the mass nor the charge of the ion.

   (b) Two isotopes of masses m_1 and m_2 cross the selector. Find the distance between the two points at which they strike the detector.

9. A straight conducting rod of length L slides without friction along two long horizontal rails a distance L apart, in a region of uniform magnetic field of magnitude B perpendicular to the plane of the rails. A source maintains a steady current i in the rod, in the sense that drives the rod along the rails. The rod has cross-sectional area A and contains n carriers per unit volume, each of charge $q > 0$, whose drift velocity relative to the rod has magnitude v_d, so that $i = nqv_dA$. At the instant considered the rod moves along the rails with speed u, in the direction of the Ampère force. Neglect the resistance of the rod and the random thermal motion of the carriers.

   (a) Find the rate at which the Ampère force $i\mathbf{L} \times \mathbf{B}$ does work on the rod.

   (b) Each carrier moves with the drift velocity along the rod and with the rod's velocity along the rails. Find the total rate at which the magnetic force does work on all the carriers in the rod, and reconcile it with your answer to (a).